

5 (a) (i) $I_1 = I_2 + I_3$ B1 [1]

(ii) $I = V / R$ or $I_2 = 12 / 10$ (= 1.2 A) C1
 $R = [1/6 + 1 / 10]^{-1}$ [total $R = 3.75 \Omega$] or $I_3 = 12 / 6$ (= 2.0 A) C1
 $I_1 = 12 / 3.75 = 3.2 \text{ A}$ or $I_1 = 1.2 + 2.0 = 3.2 \text{ A}$ A1 [3]

(iii) power = VI or I^2R or V^2 / R C1

$$x = \frac{\text{power in wire}}{\text{power in series resistors}} = \frac{I_2^2 R_w}{I_3^2 R_s} \text{ or } \frac{VI_2}{VI_3} \text{ or } \frac{V^2/R_w}{V^2/R_s}$$

C1

$$x = 12 \times 1.2 / 12 \times 2.0 = 0.6(0) \text{ allow } 3 / 5 \text{ or } 3:5$$

A1 [3]

(b) p.d. BC: $12 - 12 \times 0.4 = 7.2 \text{ (V)}$ / p.d. AC = 4.8 (V) C1
 p.d. BD: $12 - 12 \times 4 / 6 = 4.0 \text{ (V)}$ / p.d. AD = 8.0 (V) C1
 p.d. = 3.2 V A1 [3]

6 (a) extension is proportional to force / load B1 [1]